TC-SSFN: Suppressing Seasonal Noise and Enhancing Feature Discrimination for Land Cover Time Series Change Detection

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***Abstract*—Change detection using multitemporal remote sensing imagery was a critical tool for monitoring land surface dynamics. However, conventional methods often suffered from limited spatial–temporal feature integration, seasonal noise interference, and poor interpretability. To address these limitations, we proposed a temporal-consistent spectral–spatial fusion network (TC-SSFN) that jointly modeled local spatial–spectral structures and long-term temporal dynamics through four key modules: a spatial feature extraction (SFE) module for hierarchical spatial encoding, a temporal series modeling (TSM) module employing patch-based self-attention, a quarter-based seasonal encoding (QSE) module for suppressing interannual seasonal noise, and a deep supervision (DS) strategy that injected label-guided constraints into intermediate layers to enhance feature discriminability and interpretability. Experimental results across five diverse scenes in China demonstrated that TC-SSFN achieved state-of-the-art performance, with an average $F1$ score of 93.33% and IOU of 87.69%, consistently outperforming baseline methods. Ablation and clustering analyses further validated the complementary role of each module and revealed that supervision-induced feature structuring improved semantic separability. The quarter-based temporal segmentation strategy grouped multitemporal images by season and was combined with a differencing strategy. This combined approach effectively balanced sensitivity to both gradual and abrupt changes while preserving more meaningful change patterns. Overall, TC-SSFN provided a robust, interpretable, and generalizable framework for land cover change detection in complex, seasonally dynamic environments, offering practical value for large-scale monitoring and long-term remote sensing applications. The codes of our proposed TC-SSFN model are available at https://github.com/ChenyinD/TC-SSFN, and the datasets used in this study can be accessed at https://doi.org/10.5281/zenodo.18222086**

***Index Terms*—Change detection, remote sensing time series, seasonal robustness, spectral–spatial fusion, temporal consistency.**

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I. Introduction

CHANGE detection is the process of identifying spatial variations induced by human or natural phenomena through multitemporal satellite imagery [1]. With the continuous launch of satellites and the free access to remote sensing data, multitemporal remote sensing imagery covering the same geographic region has become increasingly valuable for detecting land use and land cover (LULC) changes [2]. Multitemporal change detection has become a critical tool in various domains, including land use/land cover change [3], natural disaster assessment [4], and urban expansion [5].

Compared to bitemporal change detection methods, time series change detection can more comprehensively capture long-term trends and abrupt changes in the evolution of land cover [6], [7]. Its accuracy heavily depends on the collaborative use of spatial, spectral, and temporal features [8]. Spatial features are used to characterize the structural morphology of land cover [9], [10], spectral features reflect the physical properties of land cover [11], and temporal features record the evolution process of land cover [12]. At the same time, spectral–temporal features integrate both spectral and temporal information, revealing the temporal patterns of land cover change and reflecting periodic seasonal phenological characteristics [13]. Although previous studies have demonstrated the advantages of time series methods in change detection [14], significant differences remain across methods in how they leverage spatial, spectral, and temporal features [15].

Traditional time series change detection methods such as CCDC [16], LandTrendr [17], and BFAST [6] model spectral trajectories over time by fitting parametric functions at the pixel level [18]. Specifically, CCDC employs harmonic regression to capture continuous spectral variation and detect gradual change. LandTrendr applies piecewise linear fitting to extract temporal breakpoints, while BFAST decomposes time series into seasonal, trend, and residual components to identify abrupt transitions. These methods offer strong temporal modeling capabilities and can effectively track land cover dynamics [19]. However, they operate independently

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on each pixel, neglecting spatial context [20]. As a result, they often produce fragmented or noisy change maps, lacking structural coherence across regions.

Deep learning-based approaches have advanced change detection by jointly modeling spatial, spectral, and temporal features [21]. For instance, Li et al. [22] use CNNs and ConvBiLSTM to extract spatial features and learn time-differenced representations. Saha et al. [23] employ pretrained CNNs followed by LSTMs to capture frame-level temporal patterns. Yang et al. [24] propose an unsupervised convolutional recurrent network that integrates time-distance constraints. While these models improve spatial consistency and temporal pattern recognition, they typically rely on fixed-length, cloud-free time series, which leads to temporal sparsity and information discontinuity [25].

In recent years, Transformer architectures, leveraging powerful self-attention mechanisms and strong global modeling capabilities, have effectively overcome the limitations of CNNs in capturing long-range dependencies. Furthermore, an increasing number of studies have begun to explore spatiotemporal interactions. Ma et al. [26] propose a spatiotemporal interaction Transformer architecture for multitemporal feature extraction, which incorporates frequency-domain representations to enrich spectral information for change detection. Hafner et al. [27] employ self-attention mechanisms to enhance multitemporal building representations while maintaining temporal continuity, thereby enabling more accurate detection of gradual and continuous changes. However, Transformer-based methods are often computationally intensive, which limits their scalability in large-scale change detection tasks. To address these computational challenges, the emerging Mamba architecture has demonstrated strong potential for efficient, computation-intensive change detection. Chen et al. [28] introduce three spatiotemporal modeling mechanisms seamlessly integrated with the Mamba architecture, fully leveraging its capabilities to achieve comprehensive spatiotemporal interaction modeling. Xie et al. [29] further integrate Mamba modules with Transformer components to enhance the extraction of spatial and spectral features. Although existing methods have advanced the modeling capabilities and computational efficiency of change detection, they still inadequately capture phenological patterns in long-term time series imagery and lack mechanisms to mitigate phenological interference. Consequently, seasonal pseudochanges may appear in the change detection results, reducing overall detection accuracy [30].

To address the challenges of spurious seasonal variations and feature ambiguity in remote sensing time series, this article proposes a temporal-consistent spectral–spatial fusion network (TC-SSFN). Drawing inspiration from the patch-based modeling paradigm of PatchTST [31], the TC-SSFN framework employed a novel seasonal-cycle-aligned patching strategy. This strategy decomposed the input time series into patches synchronized with natural phenological cycles. Combined with spatiotemporal–spectral collaborative feature encoding and label-guided feature disentanglement mechanisms, the proposed framework significantly enhanced the robustness and interpretability of change detection. The intrinsic alignment of temporal patches with seasonal dynamics mitigated interference from pseudochanges while improving feature coherence.

Despite their effectiveness, deep learning models have complex structures and are difficult to interpret, and the high-dimensional features extracted during training often lack clear semantics [32], [33]. When faced with remote sensing scenes with complex geographical distributions and diverse land types, the models are prone to capturing patterns that are unrelated to changes or lack discriminative power, thereby affecting the accuracy and reliability of change detection [34], [35]. Conventional change detection frameworks typically apply supervision only at the decision layer, relying on classification loss at the final output to indirectly guide feature learning [36]. While some approaches attempt to introduce auxiliary classifiers at intermediate layers, such supervision remains inherently limited in its ability to regulate the internal feature extraction process [37]. As a result, the model tends to autonomously determine which features to prioritize, which may lead to the learning of patterns that are misaligned with the task objective—particularly under complex geographical conditions and in the presence of pseudochanges. This limitation underscores the need for a more principled and task-aligned supervisory mechanism that can guide the learning of semantically meaningful and temporally robust representations throughout the network. To intrinsically mitigate this limitation, we introduced a new deep supervision (DS) mechanism that injected auxiliary signals into intermediate spatial–temporal feature layers. This strategy guided the model to learn stagewise features highly relevant to change patterns. Diverging from traditional classifier-dependent supervision paradigms [38], our lightweight label-guided approach introduced a zero-parameter constraint mechanism that directly utilized change labels (changed/unchanged) to regulate the learning dynamics of temporal features. Specifically, for unchanged locations, the model was encouraged to maintain high cross-temporal feature consistency, ensuring stability over time; whereas, for changed locations, it was guided to enhance discriminative differences between temporal features, thereby promoting more distinct and interpretable change representations.

Based on these principles, we developed TC-SSFN as a modular architecture comprising four key components. The spatial feature extraction (SFE) module used shared CNNs to extract spatial–spectral representations from each temporal snapshot. The temporal series modeling (TSM) module encoded quarterly patches using multihead self-attention-based modeling. The quarter-based seasonal encoding (QSE) computed interannual seasonal differences to suppress cyclical noise. Finally, the DS module applied label-guided constraints to intermediate outputs, reinforcing semantic alignment throughout the network. Together, these modules enabled TC-SSFN to robustly detect land cover changes across complex and seasonally dynamic environments. The main contributions of this work were summarized as follows.

1) We proposed a seasonality-aware modeling framework for time series change detection, which addressed seasonal effects through a unified design combining temporal partitioning, trend–seasonal decomposition, and interannual

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same-season differencing, enabling the model to focus on persistent land cover changes rather than seasonal variations.

2) We proposed a parameter-free DS mechanism that actively shaped the latent feature space into a structure more conducive to change separation by directly imposing label-guided consistency and dissimilarity constraints on the features. This paradigm differed from conventional DS strategies that relied on auxiliary classifiers, offering a more direct and efficient form of guidance for feature learning in change detection.

The structure of this article is organized as follows. Section II introduces the dataset used in the experiments. Section III provides a detailed description of the proposed method. Section IV presents the experimental process and analysis of results. Section V discusses the improvements to the model and the role of each module. Section VI concludes this article with a summary of the work.

II. Study Area and Material

A. Study Area

To evaluate the generalization capability of our method across different regions, we conducted experiments on six representative scenes. These study areas are located in various cities across China: 1) Hefei, Anhui (S1); 2) Nanping, Fujian (S2); 3) Yichun, Jiangxi (S3); 4) Zhengzhou, Henan (S4); 5) Nanyang, Henan (S5); and 6) Hangzhou, Zhejiang (S6). The dominant land cover types in these areas include cropland, forest, impervious surfaces, and water bodies. These study areas collectively illustrate various patterns and intensities of land cover change under urban expansion, demonstrating the robustness and general applicability of our method.

B. Research Data and Preprocessing

In this study, Landsat 8 imagery was selected as the data source for both training and testing. Within the Google Earth Engine (GEE) platform, the Landsat 8 dataset provided atmospherically corrected surface reflectance products with a spatial resolution of 30 m, making it well-suited for large-scale geospatial analysis (Amani et al., 2020). The dataset covered multiple spectral bands, including blue (0.450–0.515 μm), green (0.525–0.600 μm), red (0.630–0.680 μm), near-infrared (NIR, 0.845–0.885 μm), and shortwave infrared (SWIR 1 and SWIR 2, 1.560–1.660 and 2.100–2.300 μm, respectively) [40]. With a revisit interval of approximately 16 days, Landsat 8 offered a rich temporal resolution that supported long-term change detection and temporal dynamics analysis [41].

In this study, six different geographic scenes, each with a spatial size of 400 × 400 pixels, were selected for training and testing. These scenes encompassed various land cover types, including forest, grassland, barren land, built-up areas, cropland, and water bodies. Overall, the observed changes were largely concentrated in the mutual conversions among farmland, forest, and impervious surfaces, while the stable areas were dominated by impervious surfaces. Each scene spanned a time period from January 1, 2013, to December 31, 2023, during which multiple land cover transitions occurred. To evaluate the applicability of the proposed TC-SSFN model, change and nonchange regions were manually annotated within each scene. Specifically, a label of "0" indicated unchanged regions, "1" denoted changed regions, and "−1" represented regions that were not classified. In this study, labeled change and nonchange areas from Scene 6 were used as training data, while the remaining scenes served as validation datasets. The training dataset consists of a single scene image, comprising 11 678 labeled change samples and 9932 unchanged samples. The validation dataset includes five scene images, containing a total of 47 768 change samples and 30 929 unchanged samples. The labels for both changed and unchanged regions were obtained via visual interpretation utilizing Google Earth, and all labels corresponding to the six scenes are publicly available at https://doi.org/10.5281/zenodo.18222086

To ensure high-quality, temporally continuous, and seasonally interpretable inputs for the change detection model, we designed a three-stage preprocessing pipeline. The first step operated directly on raw Landsat 8 imagery to address cloud contamination and data availability. The second and third steps were conducted within the temporal modeling branch, where pixelwise time series were further processed to enhance smoothness and to separate trend and seasonal components for improved temporal representation.

1) Temporal Gap Filling: To fully leverage the temporal resolution advantage of Landsat 8 imagery, this study differed from conventional multitemporal approaches that typically selected only a limited number of cloud-free images [43]. Instead, all available images within the study period were incorporated into the analysis [44]. Given the inevitable presence of high-cloud-coverage observations in long-term time series, we first constructed a multiband image collection in GEE, organized sequentially by band and acquisition date. To address the issue of missing data caused by cloud contamination, we proposed a hierarchical imputation strategy. Specifically, for each pixel, a complete time series table was constructed. Missing values were first filled by grouping data by month and replacing cloud-covered observations with the mean values from the same month. If missing values remained for certain months, the mean value across the same quarter was used as a secondary replacement. As a final fallback, the global mean was employed to fill any remaining gaps. This strategy effectively preserved the continuity of the time series while mitigating noise introduced by cloud interference, thereby ensuring both data quality and completeness.

2) Temporal Interpolation: Landsat 8 remote sensing images had a temporal resolution of approximately 16 days, with around 23 images per year, forming a discrete and sparse time series [45]. To enhance the analysis of trends and seasonality in the time series, we performed interpolation on these sparse time series. We chose quadratic polynomial interpolation, which smoothed the time series curve while minimizing the introduction of noise.

3) Time Series Decomposition: To enable the model to focus more on different structures within the time series, we used a classical additive seasonal model to decompose the time series into trend and seasonal components [46].

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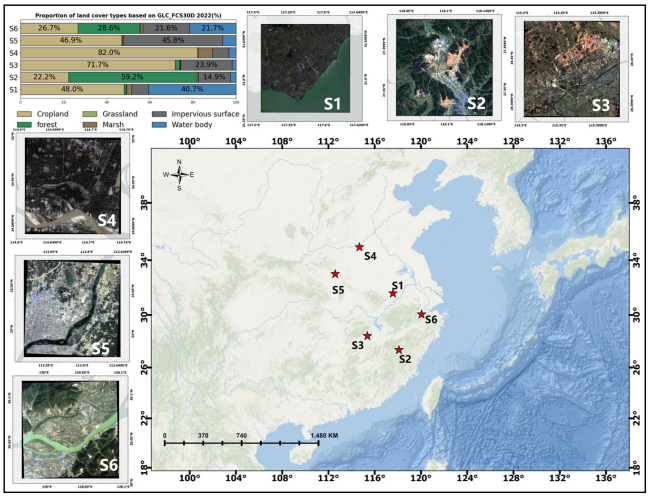


Fig. 1. Spatial positions of the six experimental scenes used in this study. S1–S6 represented selected areas within six cities: Hefei (Anhui), Nanping (Fujian), Yichun (Jiangxi Province), Zhengzhou (Henan), Nanyang (Henan), and Hangzhou (Zhejiang). The bar chart in the upper-left corner illustrated the distribution of different land cover types in these study areas based on the GLC_FCS30D [42] 2022 dataset.

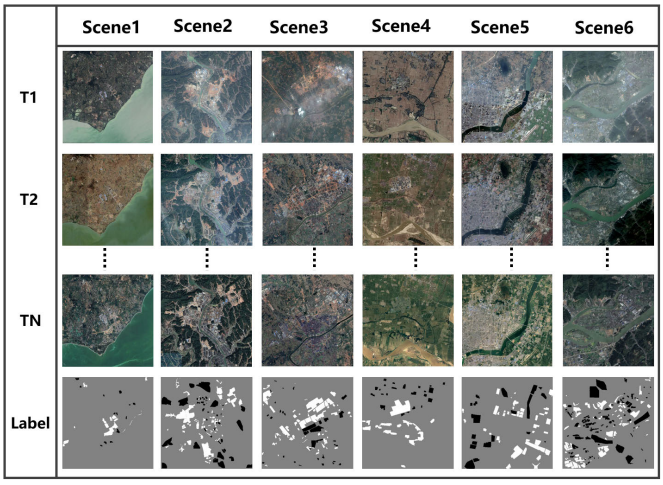


Fig. 2. Illustration of the six experimental scenes used in this study. The bottom row presented the ground truth masks, where white pixels represented changed areas, black pixels represented unchanged areas, and gray pixels represented unlabeled regions.

Specifically, the classical additive seasonal model extracted the trend component through sliding window average pooling, smoothing short-term fluctuations and enhancing long-term change patterns. Then, by removing the trend component, we extracted the residual component and the seasonal component. Assuming that the seasonal period is k, the seasonal effect for each position j within the range {1, 2, ..., k} was estimated by averaging the detrended values that correspond to the same seasonal position across multiple cycles. This approach helped to capture the recurring seasonal patterns by aligning data points from the same season across different years, thus isolating the seasonal component from the overall trend. The following was the calculation formula for the aforementioned process:

$$T_t = \frac{1}{k} \sum_{i=t-[k/2]}^{t+[k/2]} X_i \tag{1}$$

$$S_t + R_t = X_t - T_t \tag{2}$$

$$S_j = \frac{1}{N_j} \sum_{i=1}^{N_j} \left(X_{j+(i-1)k} - T_{j+(i-1)k} \right) \tag{3}$$

$$S_j^{norm} = S_j - \frac{1}{k} \sum_{i=1}^{k} S_i \tag{4}$$

where X_t represents the value of the original time series at time step t, k represents the size of the moving average window, j represents the seasonal index, T_t represents the trend component, S_t represents the seasonal component, R_t represents the residual component, and S_j^{norm} represents the seasonal component after removing the mean value.

III. Method

In this work, we proposed TC-SSFN, a supervised learning framework for binary (changed/unchanged) land cover change detection. The model took long-term remote sensing image time series as input and produced a per-pixel change probability map as output. Fig. 3 illustrates the overall architecture of the TC-SSFN proposed in this study. The original stacked remote sensing imagery was first interpolated to form a 4-D data volume with the shape [spectrum × time × width × height]. To fully exploit the spatial, spectral, and temporal information embedded in remote sensing data, the model extracted features in parallel through the SFE module and the temporal series modeling (TSM) module. This dual-stream parallel architecture was designed to enable SFE and temporal modeling to learn their respective patterns independently, thereby reducing interference between spatial and temporal representations. The decoupled design also facilitated separate optimization of each module and provided a clear structural basis for applying DS to enhance the discriminability of both spatial and temporal features. In the SFE, a convolutional neural network with shared parameters was applied independently to each time slice. Through a series of convolutional layers, local spatial–spectral features were progressively aggregated toward the central pixel, effectively capturing spatial context and spectral dependencies. In the TSM, the module constructed a spectral–temporal curve centered on the target pixel. It incorporated a trend-seasonal decomposition technique alongside a multihead self-attention mechanism applied across both temporal and spectral dimensions, allowing for the extraction of deep spectral–temporal representations. To more effectively capture recurring seasonal patterns, QSE Modules were strategically embedded at multiple locations within the network architecture. These modules were designed to mitigate the influence of seasonal fluctuations, thereby improving the model's sensitivity to meaningful changes and its overall robustness. The extracted difference features were further constrained through DS, which enhanced the balance and interpretability of the features derived from spatial and temporal dimensions.

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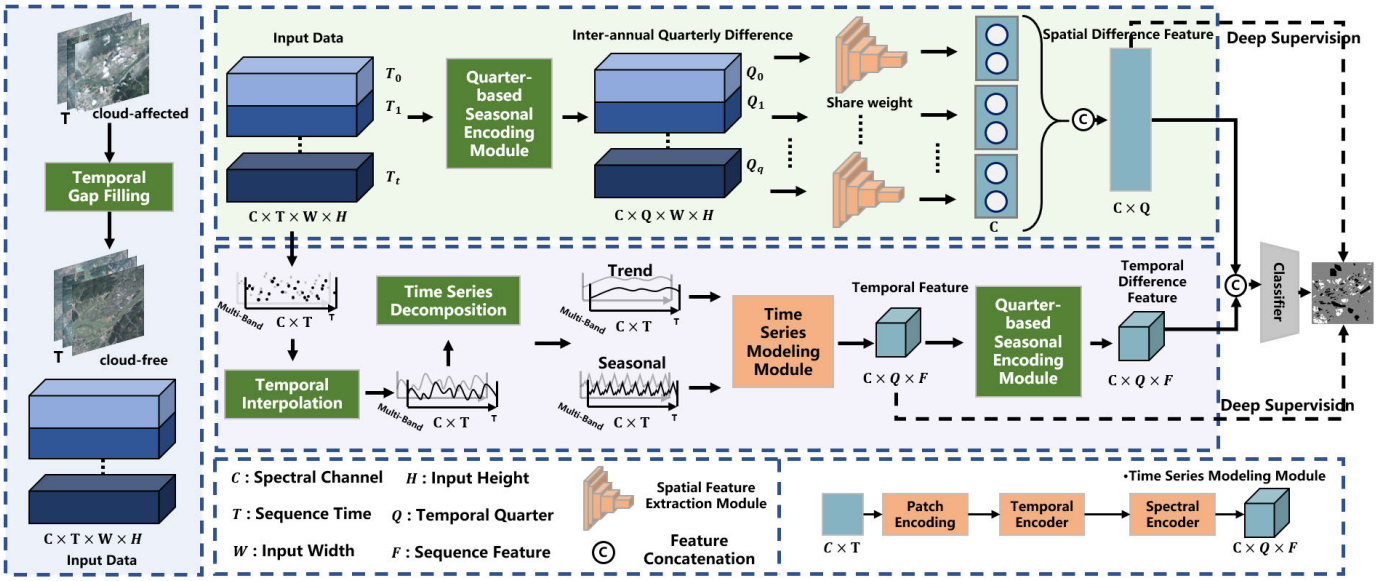


Fig. 3. Overview of the TC-SSFN architecture. TC-SSFN consisted of four modules: the SFE module extracted spatial–spectral features from each snapshot using shared CNNs, the time series modeling module modeled quarterly sequences via multihead self-attention, the QSE module computed interannual seasonal differences to suppress periodic noise, and DS applied label-guided supervision to intermediate outputs. In the diagram, the green portion represented nonparametric operations, while the orange portion represented parametric training.

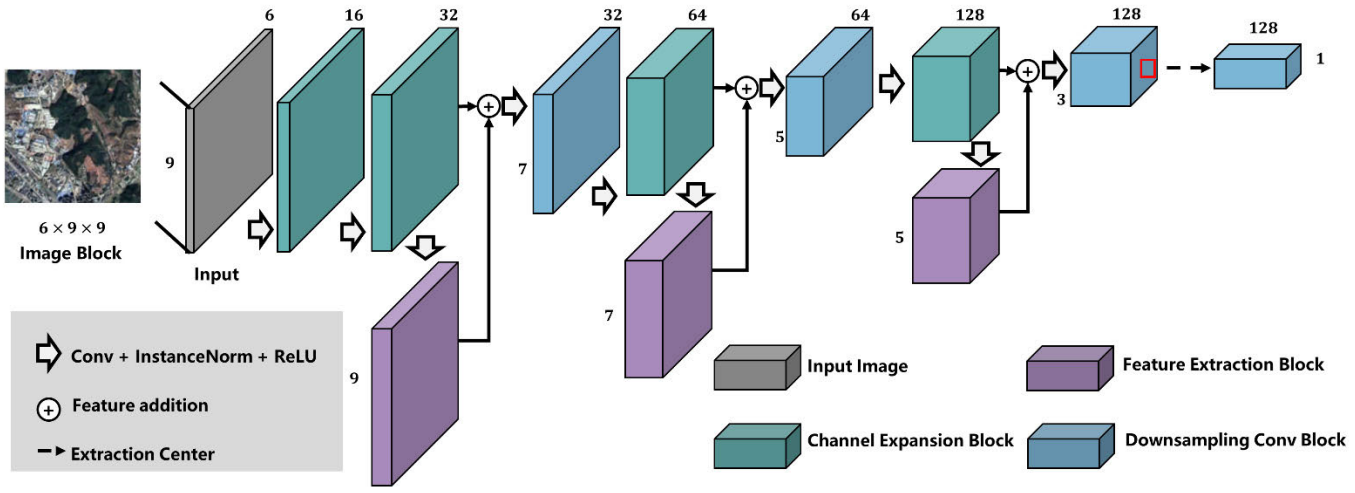


Fig. 4. Structure of the SFE module. A multilayer convolutional neural network was utilized to extract and hierarchically aggregate spatial features, enabling the progressive learning of spatial representations.

The detailed architectures and functionalities of each component shown in Fig. 3 are discussed in Sections III-A–III-D.

A. SFE Module

In the SFE module, we designed a series of convolutional blocks to progressively extract and compress spatial features from local image patches, aiming to obtain an efficient and context-aware representation for each pixel. As illustrated in Fig. 4, this module aggregated spatial information surrounding the central pixel step by step through multiple convolutional layers, ensuring that the extracted features comprehensively captured the spatial distribution of the input image. Specifically, for each pixel, a 9×9 neighborhood patch was extracted as the input. A 9×9 patch size was selected to ensure that the convolutional layers' receptive field covered the full spatial extent of each local region, allowing the model to capture both fine-grained textures and broader contextual cues around the target pixel [47]. This patch-based approach processed the image in a pixel-centered manner, with each output feature representing the central pixel of its corresponding patch, thereby preserving spatial precision while incorporating local context. The 9×9 patch was gradually downsampled through three consecutive convolutional layers with a kernel size of 3×3, stride = 1, and no padding, reducing the feature map size from 9×9 to 3×3. The central feature vector of the resulting 3×3 map was then selected as the output representation of the input pixel. This design emphasizes the central region while mitigating the influence of heterogeneous surroundings, which is particularly beneficial in heterogeneous land cover

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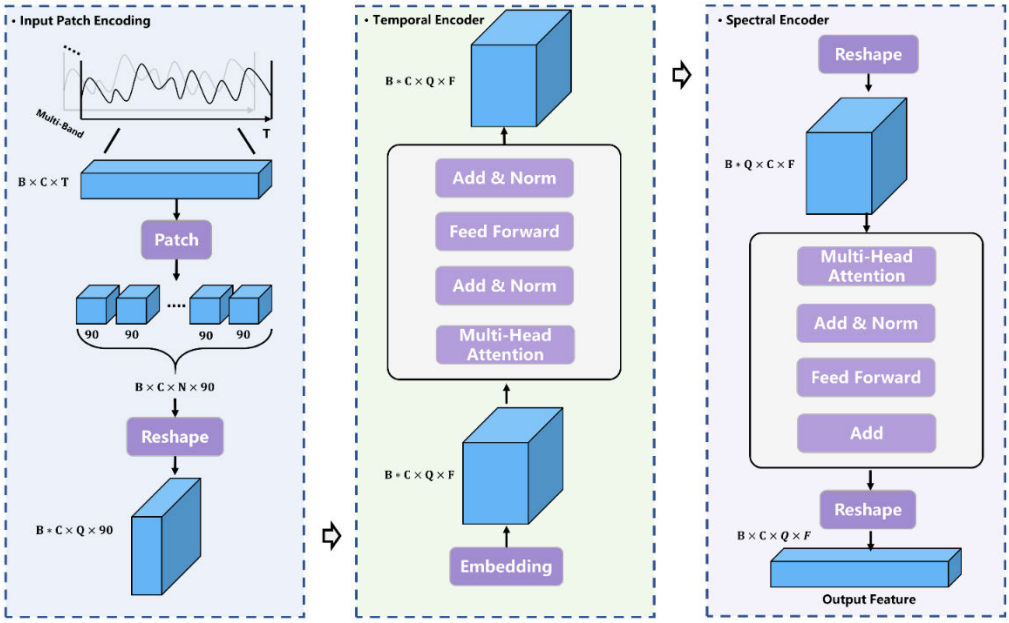


Fig. 5. Structure of the temporal series modeling module. This module integrates multihead self-attention mechanisms applied along both the spectral and temporal dimensions to effectively capture joint spectral–temporal dependencies.

scenarios. Throughout this convolutional process, each layer captured increasingly abstract spatial patterns, and their hierarchical integration enabled the learning of semantically rich and scale-aware representations. To further enhance feature expressiveness and alleviate information loss caused by spatial compression, we progressively increased the number of channels across layers [16, 32, 64, 128], thereby strengthening the model's representational capacity while maintaining critical spatial information.

B. Temporal Series Modeling Module

The TSM consisted of three key components: nonoverlapping sliding window segmentation, a patch-group-based multihead self-attention mechanism, and a spectral-based multihead self-attention mechanism. As illustrated in Fig. 5, the time series was divided into local patches using a fixed-size (90) nonoverlapping window, effectively reshaping the temporal dimension into a window dimension and a group dimension. This segmentation strategy preserved the continuity of temporal structure while enabling the model to capture stable temporal features across multiple scales, thereby facilitating the representation of seasonal variations and trend patterns.

Subsequently, to explore the dependencies between different patch groups, an initial multihead self-attention module was applied for temporal modeling. Following common practice in Transformer-based sequence modeling, each spectral channel was initially processed independently to capture temporal dependencies within individual bands. This module comprised several essential components, including input embedding, positional encoding, multihead self-attention, feed-forward neural networks, residual connections, and layer normalization. First, the input sequences were projected into a high-dimensional feature space through an embedding layer, while positional encoding was incorporated to inject temporal order information, compensating for the sequence-agnostic nature of the self-attention mechanism. The multihead self-attention further captured long-range dependencies within the sequence. Residual connections and layer normalization enhanced training stability, and the feed-forward network extracted deeper and more nonlinear representations.

Following the channel-independent temporal modeling, we further established relationships across spectral bands. Specifically, a spectral-based multihead self-attention mechanism was introduced within the TSM, where features were organized along the patch group dimension. This design enabled the model to capture interband dependencies and semantic relationships based on the learned temporal representations, resulting in joint modeling of temporal and spectral information.

C. QSE Module

To reduce pseudochanges caused by seasonal variation, we adopted a strategy of comparing the same quarters across adjacent years to transform the image sequence into a difference sequence. This allowed the model to more intuitively understand the nature of actual changes.

In the temporal–spectral feature construction, we extracted the feature representation and compared it with the same quarters in adjacent years. By computing the channelwise absolute differences between the current quarter and these cross-year same-quarter neighbors and averaging across all neighbors, we obtained a difference score that reflected the intensity of change at that time step. A higher difference score indicated a greater deviation from the temporal context, suggesting a higher likelihood of real change. This approach incorporated local temporal context constraints to enhance the

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model's ability to perceive true changes while improving its robustness against seasonal variations.

For spatial information extraction, we grouped the data by year and quarter, and computed the average of each group to represent the spatial characteristics of that quarter. This process not only incorporated temporal information into SFE but also smoothed temporal noise to some extent. Subsequently, we calculated the differences in spatial features between the same quarters of adjacent years. This interannual differencing approach effectively captured long-term trend changes while significantly reducing the impact of seasonal fluctuations. By eliminating periodic noise, the model was able to better focus on long-term dynamic changes, thereby improving the accuracy of change detection in regions of interest. The detailed computation is given as follows:

$$Year_t = X_{mark}[t, 0]$$
$$Month_t = X_{mark}[t, 1]$$
$$Quarter_t = \left[\frac{Month_t - 1}{3}\right] + 1 \tag{5}$$
$$Mask_{y,q} = \left\{t | Year_t = y, Quarter_t = q\right\} \tag{6}$$
$$Diff_{y,q} = Avg_{y,q} - Avg_{y-1,q} \tag{7}$$
$$T_{stack} = Stack\left(Diff_{y,q}\right) \tag{8}$$

where X_{mark} represents the year, month, and time of the time series, $Avg_{y,q}$ represents the average feature value in y year q quarter, and $Stack$ represents stacking the quarterly feature differences in the time dimension.

The proposed QSE module embedded a structured, seasonally aligned differencing operation directly into the network. This encoded a key phenological prior: meaningful change should have been assessed by comparing the same seasonal phase across years. By constructing interannual differential signals for identical quarters, the module directly suppressed cyclical fluctuations, allowing subsequent layers to focus on learning discriminative features for genuine change rather than inferring seasonal invariance from data alone.

D. Deep Supervision

To make feature learning more semantically meaningful and interpretable, we introduced a novel dual intrinsic guidance mechanism via auxiliary supervision applied directly to the outputs of both the SFE and the TSM. This approach encoded the core priors of change detection—temporal consistency and spatial stability for unchanged pixels versus feature divergence for changed pixels—into the feature learning process itself, guiding the model to form a more separable and task-aligned latent representation.

The SFE outputs spatial difference features centered and stacked over time, where each time step represents the spatial difference between the same quarter in the current year and its adjacent years. To directly supervise this spatial pathway, we formulated a spatial change intensity score. When a change occurred, we expected this difference to be amplified; otherwise, it should ideally approach zero. To quantify this, we computed the mean of the absolute spatial difference features (of shape [Batch, Channel, Quarter]) over the channel and quarter dimensions and applied a Tanh activation to scale the results into the [0, 1] range, serving as a predicted and interpretable change indicator

$$Out_{SFE} = Tanh\left(\frac{1}{C * Q}\sum_{c=1}^{C}\sum_{q=1}^{Q}|In_{SFE}|\right) \tag{9}$$

where In_{SFE} represents the input spatial features, C represents the channel dimension, Q represents the seasonal dimension, and Tanh represents the hyperbolic tangent activation function.

The temporal sequence module (TSM) generated seasonal groupings of temporal features. To supervise its inherent sequential modeling, we design a temporal deviation intensity score. We extracted the features from the first and last quarters and computed their average differences with features from the same seasonal group across the sequence. The resulting difference was normalized using the Tanh function. This process is designed to align with the change labels, aiming to maximize the separability of change features while promoting the compactness of unchanged features in the feature space, thereby directly guiding the module's sensitivity to temporal anomalies

$$Out_{TSM} = \text{Tanh}\left(\frac{1}{Q-1}\sum_{q=1}^{Q-1}\left(\frac{1}{C}\sum_{c=1}^{C}\left|In_{extremity} - In_{mid}\right|\right)\right) \tag{10}$$

where $In_{extremity}$ represents the feature representation of the first quarter and the last quarter in the temporal sequence and In_{mid} represents intermediate features that correspond to the same seasonal group as either the first or last quarter. Tanh represents the hyperbolic tangent activation function, C represents the channel dimension, and Q represents the quarter dimension.

Finally, the predicted change intensity scores produced by the SFE and the TSM, respectively, were supervised using mean squared error (mse) with respect to the ground-truth binary labels as auxiliary objectives. The primary optimization objective of the network was defined by a binary cross-entropy (BCE) loss applied to the final change detection output. The overall training objective was formulated as a weighted sum of the primary BCE loss and the two auxiliary mse losses. Specifically, each auxiliary loss associated with the SFE and TSM branches was assigned a weight of 0.5, while the BCE loss served as the dominant supervisory signal. The auxiliary supervision was introduced to provide a more robust optimization direction for intermediate representations while still allowing the primary loss to dominate the overall learning process. This loss formulation enabled the auxiliary supervision to regularize intermediate feature representations without overwhelming the optimization of the primary detection objective. By jointly optimizing the final prediction and the intermediate change intensity representations, the network was guided toward learning discriminative representations for changed regions while maintaining stable representations for unchanged regions, thereby improving sensitivity to

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meaningful spatiotemporal changes and enhancing the interpretability of the learned features

$$L_{total} = L_{main} + 0.5 * L_{SFE} + 0.5 * L_{TSM}. \tag{11}$$

This dual-guidance design introduced a direct learning objective for the feature extraction process itself, moving beyond passive feature transformation. By imposing physics-driven constraints—enforcing temporal consistency for unchanged pixels and encouraging feature divergence for changed pixels—the module actively shaped the latent feature space to be more separable with respect to the final change detection task. This internal guidance mechanism complemented the primary classification loss and ensured that more discriminative and semantically aligned representations were learned throughout the network's hierarchy.

IV. Experimental Results and Analysis

A. Experimental Setup

In the experimental setup, all tests were conducted on a GeForce Quadro 4000 GPU, and the deep learning models were implemented using the PyTorch framework. During training, the Adam optimizer was employed with parameters $\beta_1 = 0.9$, $\beta_2 = 0.99$, a weight decay of 0.0001, and an initial learning rate of 0.0001, which was gradually decayed by a factor of 0.86, and the total number of training epochs was set to 15. The CrossEntropyLoss function was used as the training objective, while the MSELoss function was used for DS. This configuration ensured stable convergence while maintaining a balance between training efficiency and model performance.

B. Evaluating Indicator

To comprehensively evaluate the performance of the model, multiple evaluation metrics were employed, including macro-precision (M-PRE), macro-recall (M-REC), macro-$F1$ (M-$F1$), and macro-intersection over union (M-IOU). These metrics reflected the model's performance in change detection tasks from different perspectives. Since it was challenging to annotate every region of the large-scale remote sensing images with precise change or no-change labels, we adopted macroaveraged metrics to ensure a fair and unbiased comparison of model performance, particularly under conditions of spatial annotation sparsity and potential class imbalance. The specific formulations are defined as follows:

$$\begin{aligned} PRE &= \frac{TP}{TP+FP}, M-PRE \\ &= \frac{PRE_{Change} + PRE_{Unchange}}{2} \end{aligned} \tag{12}$$

$$\begin{aligned} REC &= \frac{TP}{TP+FN}, M-REC \\ &= \frac{REC_{Change} + REC_{Unchange}}{2} \end{aligned} \tag{13}$$

$$\begin{aligned} F1 &= \frac{2 \times PRE \times REC}{PRE+REC}, M-F1 \\ &= \frac{F1_{Change} + F1_{Unchange}}{2} \end{aligned} \tag{14}$$

$$\begin{aligned} IOU &= \frac{TP}{TP+FP+FN}, M-IOU \\ &= \frac{IOU_{Change} + IOU_{Unchange}}{2} \end{aligned} \tag{15}$$

where TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative in change detection. $F1_{Change}$ and $F1_{Unchange}$ represent the $F1$ scores of the "change" and "no-change" classes.

C. Comparison With Benchmark Methods

We conducted comparative experiments between TC-SSFN and four state-of-the-art time series change detection methods across five representative scenes. For CCDC and LandTrendr, evaluations were performed directly on the GEE platform using the recommended parameters and the original Landsat 8 imagery. In contrast, UTRNet, L-Unet, MsFNet, ContUrbanCD, and our proposed method all involved preprocessing of the Landsat image collections prior to model training and evaluation. They were given as follows.

1) LandTrendr [17]: A time series analysis method based on segmented linear modeling. It performed piecewise linear fitting on vegetation indices such as NDVI to detect breakpoints within long-term change trajectories, thereby extracting the magnitude and direction of annual changes over time.

2) CCDC [16]: A time series regression-based change detection method. It constructs per-pixel regression models using continuous time series data to automatically fit seasonal trends, enabling precise identification of change timing when deviations from the modeled pattern occur.

3) L-UNet [48]: A deep multitask network based on a fully convolutional LSTM architecture. By embedding ConvLSTM modules at multiple levels of the UNet encoder, it enabled joint modeling of spatial and temporal features.

4) UTRNet [24]: An unsupervised time-distance-guided convolutional recurrent network. This model leveraged irregular temporal intervals as guidance to mitigate the impact of pseudochanges, effectively enhancing the model's sensitivity to genuine temporal variations while suppressing noise introduced by inconsistent observation timing.

5) MsFNet [49]: A multiscale feature fusion module further combined deep feature representations, preserving fine-grained spatial details, and enabling more accurate and robust change detection across diverse real-world datasets.

6) ContUrbanCD [27]: A Transformer-based feature refinement module effectively captured temporal dependencies while maintaining the sequence structure, enabling consistent and accurate detection of urban changes across high-resolution time series.

As shown in Table I, TC-SSFN outperformed the other methods, demonstrating its superior performance and robustness in change detection tasks. Specifically, the proposed TC-SSFN model achieved an average $F1$ score of 93.33% across five representative scenes, significantly surpassing other baseline methods. This indicated that TC-SSFN was more accurate and stable in capturing land cover changes. Moreover, the model attained an average IOU of 87.69%, reflecting its superior spatial consistency and remarkable capability in precisely localizing change regions.

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TABLE I
COMPARISON OF INDICATORS

Scene	Method	M-PRE(%)	M-REC(%)	M-F1(%)	M-IOU(%)
Scene1	LandTrendr	66.87	67.88	51.57	34.75
	CCDC	81.70	86.01	83.39	72.22
	L-Unet	56.93	58.02	57.11	42.37
	UTRNet	68.43	71.49	69.40	54.69
	MsFNet	75.12	70.14	69.46	53.75
	ContUrbanCD	70.76	69.40	69.33	53.29
	TC-SSFN	**96.60**	**95.41**	**95.99**	**92.36**
Scene2	LandTrendr	83.09	74.77	72.96	57.87
	CCDC	88.64	88.64	88.64	79.60
	L-Unet	89.50	88.97	89.03	80.24
	UTRNet	88.26	85.23	85.19	74.27
	MsFNet	93.10	88.71	90.56	83.05
	ContUrbanCD	91.54	91.52	91.53	84.38
	TC-SSFN	**93.53**	**92.94**	**93.02**	**86.96**
Scene3	LandTrendr	66.94	62.07	45.37	29.54
	CCDC	92.46	79.76	85.65	74.89
	L-Unet	77.43	81.77	78.59	65.28
	UTRNet	59.53	50.71	63.99	36.80
	MsFNet	83.81	67.53	69.99	56.78
	ContUrbanCD	90.22	73.19	76.75	64.23
	TC-SSFN	**94.79**	**88.29**	**90.85**	**83.54**
Scene4	LandTrendr	57.42	53.44	31.35	19.08
	CCDC	72.56	79.30	72.29	57.23
	L-Unet	65.21	65.40	50.64	33.90
	UTRNet	67.12	51.63	46.54	39.10
	MsFNet	86.99	82.79	82.59	70.47
	ContUrbanCD	89.50	86.76	86.80	76.73
	TC-SSFN	**90.06**	**92.93**	**91.30**	**84.23**
Scene5	LandTrendr	70.84	58.01	48.36	33.82
	CCDC	74.08	68.54	65.03	48.59
	L-Unet	65.44	64.37	64.17	47.52
	UTRNet	83.67	80.21	80.36	67.34
	MsFNet	91.12	81.98	85.30	75.32
	ContUrbanCD	87.36	80.02	82.77	71.80
	TC-SSFN	**95.43**	**95.54**	**95.47**	**91.34**
Average	LandTrendr	69.03	63.23	49.92	35.01
	CCDC	81.89	80.45	79.00	66.51
	L-Unet	70.90	71.71	67.91	53.86
	UTRNet	73.40	67.85	69.10	54.44
	MsFNet	86.03	78.23	79.58	67.87
	ContUrbanCD	85.88	80.18	81.44	70.09
	TC-SSFN	**94.08**	**93.02**	**93.33**	**87.69**

In contrast, the traditional LandTrendr method consistently exhibited the lowest performance across all evaluation metrics, with an average $F1$ score of only 49.92% and a mean IOU of 35.01%. The CCDC method demonstrated more balanced performance, achieving an average $F1$ score of 79.00% and a mean IOU of 66.51%, representing the strongest results among the baseline approaches. Nonetheless, it remained inferior to TC-SSFN in both accuracy and spatial consistency. Deep learning-based methods, including L-Unet, UTRNet, MsFNet, and ContUrbanCD, achieved moderate average $F1$ scores of 67.91%, 65.10%, 79.58%, and 81.44%, respectively. However, their performance exhibited considerable variability across different scenes, indicating limitations in robustness and generalization capability. In comparison, TC-SSFN not only outperformed all baseline methods across all metrics but also demonstrated the lowest interscene performance variance, highlighting its superior stability and adaptability in diverse change detection scenarios.

The competitive performance of the traditional CCDC method in certain scenes highlighted the complementary strengths and limitations of different methodological paradigms. Its pixelwise harmonic regression-based temporal modeling effectively captured spectral trajectory anomalies in homogeneous regions, but the lack of spatial contextual modeling often resulted in fragmented "salt-and-pepper" noise. In contrast, deep learning methods alleviated this limitation through rich semantic feature learning. However, many relied on implicit, data-driven mechanisms that failed to stably encode phenological priors or provide interpretable learning signals. The superior and stable performance of TC-SSFN stemmed from its principled integration of these two paradigms. The seasonally aligned design and QSE module explicitly internalized the same-season comparison logic, enabling stable suppression of seasonal variations. Meanwhile, the parameter-free DS imposed label-guided, physically meaningful constraints to regularize the feature space. By combining explicit prior modeling with deep spatial representation learning in a dual-stream architecture, TC-SSFN provided an effective and robust framework for change detection.

Fig. 6 presents a qualitative comparison of change detection results obtained by different methods. To evaluate the detection capabilities of these methods, we selected five representative land cover change scenarios across five distinct scenes. Scene 1 involved a transformation from tidal flat to vegetated zone; Scene 2 represented the conversion of bare land into buildings; Scene 3 was a case of densely packed buildings with no actual change; Scene 4 depicted a transition from farmland to construction land; and Scene 5 illustrated a change from farmland to bare land and eventually into a park.

In Scene 1, TC-SSFN, MsFNet, and ContUrbanCd accurately identified the emergence of narrow vegetated belts along the coast, with continuous boundaries and minimal false positives. In contrast, the other methods exhibited varying degrees of omission and lacked stability. Scene 2 featured the appearance of new buildings due to urban expansion. TC-SSFN demonstrated superior performance in terms of detection coverage, boundary delineation, and false positive suppression. In comparison, L-Unet, UTRNet, MsFNet, and ContUrbanCD were sensitive to shadows and surface color variations, which led to substantial misclassifications, while LandTrendr continued to suffer from significant omissions. In the unchanged Scene 3, despite the presence of complex texture patterns, TC-SSFN consistently maintained a correct prediction of no change, significantly outperforming L-Unet and UTRNet, which generated extensive false positives. This highlighted TC-SSFN's robustness against pseudochanges. In Scenes 4 and 5, where farmland transitioned into other land cover types, TC-SSFN effectively adapted to structurally complex changes, accurately extracted change regions, and preserved the completeness and clarity of detected boundaries, while MsFNet showed relatively better performance than other comparison models but still fell short of the results achieved by TC-SSFN. Traditional methods, on the other hand, exhibited fragmented results and responded weakly to such structured transformations. Overall, TC-SSFN demonstrated clear advantages in handling complex textures, capturing fine-grained

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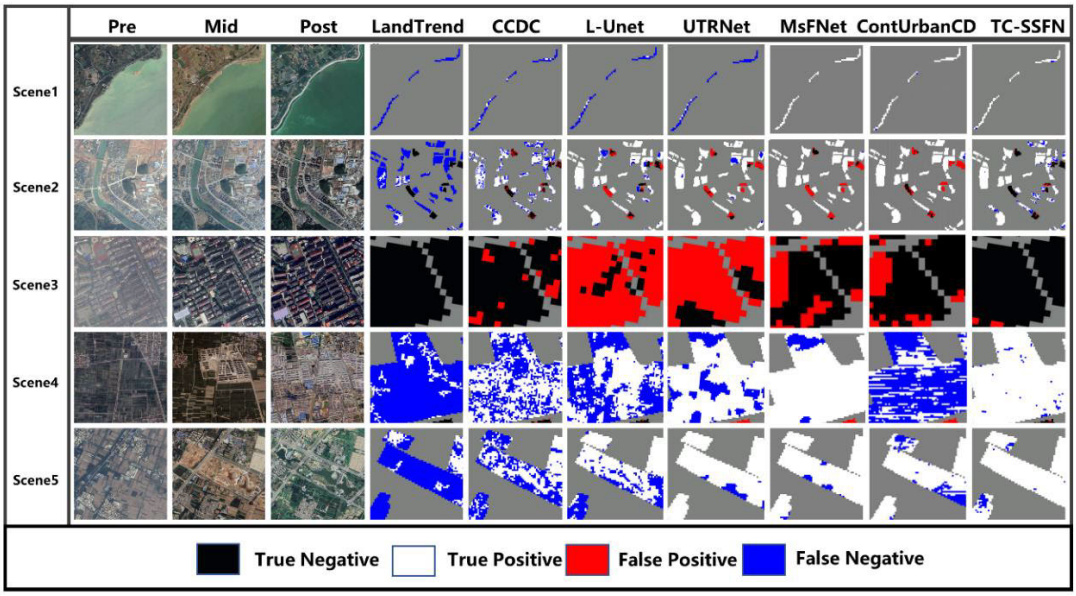


Fig. 6. Qualitative comparison of change detection results across five typical scenarios. TC-SSFN outperformed other methods: it detected narrow vegetated belts (scene 1) and new buildings (scene 2), identified no change under complex textures (scene 3), and extracted farmland transitions (scenes 4 and 5). L-Unet and UTRNet produced extensive false positives; LandTrendr exhibited substantial omissions.

change patterns, and maintaining boundary consistency. Its robustness and accuracy consistently surpassed those of existing methods, validating its effectiveness and advancement in multitemporal remote sensing change detection. The overall image result can be found in Fig. S1 in the supplementary.

D. Ablation Study

1) Componentwise Differencing Strategy Analysis: To assess the contribution of each component, we conducted ablation experiments on the SFE module, the temporal series modeling (TSM) module, the QSE module, time series decomposition (TSD), and DS. Results (see Table II) highlighted the complementary and synergistic effects of these modules on change detection accuracy. Their impact varied across scenes due to differing land cover dynamics: Scenes 1–3 were dominated by paddy field and forest conversions to bare land or impervious surfaces; Scene 4 primarily involved mutual transformations between plain farmlands and impervious surfaces; and Scene 5 exhibited complex dynamics, including temporary reurbanization—where impervious surfaces briefly converted to bare land before reconstruction—and farmland-to-impervious surface conversions.

The SFE module provided a strong spatial baseline but was insensitive to temporal variations, whereas the TSM module struggled without spatial context, leading to unstable and poorly generalized performance. Combining SFE and TSM achieved a balanced synergy between spatial structure parsing and temporal dynamic modeling, enhancing change discrimination. Although this configuration remained suboptimal without QSE and DS, it consistently outperformed single-branch setups across all scenes, demonstrating the effectiveness of fundamental spatiotemporal integration and serving as a key intermediate configuration within the full framework.

The QSE module yielded notable improvements in both spatial and temporal branches in most cases. By leveraging seasonal alignment and differencing, it effectively emphasized disruptions in temporal consistency, thereby enhancing the representation of change-related features. The improvement was especially pronounced when QSE was integrated with the SFE, suggesting that spatial representations benefited from temporally informed priors. However, in Scene 4, the temporal branch exhibited reduced accuracy, which could be attributed to the predominance of farmland areas characterized by inherently unstable interannual dynamics. In such regions, phenological variations driven by crop rotation, management practices, and interannual climate variability often produce substantial temporal deviations even in the absence of actual land cover change. As a result, the seasonal alignment and differencing operations in QSE tended to attenuate these high-magnitude but semantically nondiscriminative temporal variations, effectively smoothing out not only seasonal phase shifts but also part of the genuine change-related signals. This overnormalization reduced the contrast between true land cover transitions and strong yet recurrent agricultural dynamics, thereby weakening the discriminative capacity of the temporal branch in this specific scene.

Although TSD was merely a data preprocessing step, it had a significant impact on the temporal feature extraction capability of the subsequent model. Experimental results demonstrated that incorporating TSD led to more robust performance across all test scenarios, validating its effectiveness in enhancing the stability of temporal modeling.

Upon introducing the DS module, differences between early, middle, and late temporal segments were employed as a constraint loss. The DS module was fundamentally designed to amplify genuine change features while suppressing invariant ones, thereby promoting a more structured distinction between

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TABLE II
RESULTS OF ABLATION EXPERIMENTS

Scene	SFE	TSM	QSE	DS	TSD	M-PRE(%)	M-REC(%)	M-F1(%)	M-IOU(%)
Scene1	√					81.09	72.37	75.09	62.37
	√		√			94.33	92.88	93.57	88.10
	√		√	√		95.54	91.75	93.44	87.90
		√			√	62.27	57.21	35.69	22.01
		√	√		√	78.88	81.26	79.91	67.57
		√	√	√	√	63.47	58.25	36.90	22.88
	√	√			√	81.15	77.57	79.07	66.85
	√	√	√		√	94.84	94.72	94.78	90.20
	√	√	√	√		92.61	91.83	92.21	85.80
	√	√	√	√	√	**96.60**	**95.41**	**95.99**	**92.36**
Scene2	√					89.80	88.20	88.26	79.02
	√		√			91.40	91.00	91.06	83.60
	√		√	√		92.79	91.98	92.07	85.31
		√			√	68.84	63.48	60.27	43.89
		√	√		√	86.37	86.37	86.37	76.01
		√	√	√	√	76.84	75.66	75.19	60.30
	√	√			√	90.50	88.85	88.92	80.07
	√	√	√		√	92.24	91.74	91.81	84.86
	√	√	√	√		93.01	92.30	92.38	85.85
	√	√	√	√	√	**93.53**	**92.94**	**93.02**	**86.96**
Scene3	√					80.18	62.23	63.21	50.44
	√		√			92.49	83.91	86.93	77.51
	√		√	√		94.01	85.89	88.86	80.42
		√			√	69.89	73.14	70.54	55.34
		√	√		√	77.75	73.48	75.03	61.60
		√	√	√	√	78.63	79.46	79.02	66.17
	√	√			√	80.37	64.18	65.81	52.69
	√	√	√		√	92.44	82.24	85.59	75.57
	√	√	√	√		93.22	**88.71**	90.61	83.13
	√	√	√	√	√	**94.79**	88.29	**90.85**	**83.54**
Scene4	√					62.33	55.23	54.46	43.34
	√		√			73.20	80.05	70.63	54.97
	√		√	√		82.59	89.25	84.51	73.61
		√			√	62.05	56.54	56.56	44.54
		√	√		√	55.66	57.33	53.50	37.46
		√	√	√	√	60.91	64.05	59.74	43.56
	√	√			√	57.69	57.54	57.61	43.47
	√	√	√		√	74.94	82.36	73.87	59.00
	√	√	√	√		79.81	82.41	80.91	68.78
	√	√	√	√	√	**90.06**	**92.83**	**91.30**	**84.23**
Scene5	√					74.74	66.87	65.11	49.26
	√		√			91.17	91.25	91.20	83.82
	√		√	√		93.80	93.54	93.64	88.04
		√			√	68.95	68.51	67.85	51.37
		√	√		√	76.91	77.01	76.87	62.43
		√	√	√	√	82.34	82.45	82.36	70.02
	√	√			√	74.00	66.66	64.98	49.08
	√	√	√		√	90.65	90.65	90.65	82.90
	√	√	√	√		89.22	87.84	88.39	79.30
	√	√	√	√	√	**95.43**	**95.54**	**95.47**	**91.34**
Average	√					77.63	68.98	69.23	56.89
	√		√			88.52	87.82	86.68	77.60
	√		√	√		91.75	90.48	90.50	83.06
		√			√	66.40	63.78	58.18	43.43
		√	√		√	75.11	75.09	74.34	61.01
		√	√	√	√	72.44	71.97	66.64	52.59
	√	√			√	76.74	70.96	71.28	58.43
	√	√	√		√	89.02	88.34	87.34	78.51
	√	√	√	√		89.57	88.62	88.90	80.57
	√	√	√	√	√	**94.08**	**93.00**	**93.33**	**87.69**

changed and unchanged patterns. When combined with the QSE module, it effectively compensated for the oversuppression of change signals. This configuration excelled at capturing spatial structures although temporal feature representations remained somewhat unstable.

Ultimately, the model achieved its best performance when all modules—SFE, TSM, QSE, TSD, and DS—were integrated, highlighting that robust change detection depends on the synergistic combination of fine-grained spatial details, long-term temporal modeling, seasonal invariance, and multilevel supervision.

Visual comparisons in Fig. 7 provided additional insight into the role of each module across diverse land cover change scenarios. In the SFE module, the first and third rows depicted transitions from cropland to lower-reflectance land cover types, which the module consistently identified as changes. In contrast, forests in the second row were classified as unchanged, and the conversion of impervious surfaces surrounded by vegetation to cropland in the fourth row was also considered unchanged. These observations suggested that the SFE module did not fully capture the intrinsic nature of land cover change; instead, it exhibited a heuristic bias, tending to label impervious surfaces as changed while classifying vegetation as unchanged.

For the TSM module, changes from cropland to bare land in the first and second rows were not effectively detected, whereas its performance improved slightly in the third and fourth rows, which involved cropland-to-impervious surfaces transitions. Nonetheless, it still failed to extract highly discriminative change features. Overall, the two modules relied on spatial and temporal representations, respectively, to distinguish between land cover categories. When the SFE and TSM modules were combined, the resulting detections largely resembled those of the SFE module, suggesting that spatial features played a dominant role in the joint model's decision-making.

With the introduction of the QSE module, the misclassification of unchanged impervious surfaces as changes in the SFE module was effectively corrected. This improvement resulted from QSE performing seasonal differencing prior to convolution, which suppressed pseudochange responses. However, in the fourth-row scenario—where impervious surfaces surrounded by vegetation were converted into cropland—the differencing operation also reduced the contrast between spectrally similar land cover types, slightly weakening discrimination. For the TSM module, QSE helped reduce missed detections in vegetation-to-bare land transitions. This benefit primarily stemmed from QSE's design, which extracted temporal representations before applying seasonal differencing, thereby enhancing interclass separability in temporal dynamics. Nevertheless, this effect remained somewhat unstable.

With the introduction of the DS module, the model not only retained the corrective benefits provided by QSE for both SFE and TSM but also mitigated the loss of interclass contrast induced by differencing. By amplifying feature discrepancies, DS enabled the establishment of clearer decision boundaries among different land cover types. Consequently, the joint integration of all modules yielded the highest overall performance across all four land cover change scenarios. The overall image result can be found in Fig. S2 in the supplementary.

2) Effectiveness of QSE in Mitigating Seasonal Pseudochange: To evaluate the robustness of the QSE module against seasonal pseudochanges, we conducted experiments on three representative invariant land cover types—cropland, forest, and impervious surfaces—using the prediction accuracy of unchanged areas as the evaluation metric. As reported in Table III, the baseline spatial branch (SFE) achieved accuracies of 64.92%, 97.11%, and 39.60% across the three scenes, respectively. With the introduction of the QSE module, which applied seasonal differencing to suppress periodic variations, accuracies substantially increased to 96.76%, 99.58%, and 91.62%. Likewise, the temporal branch (TSM) exhibited very

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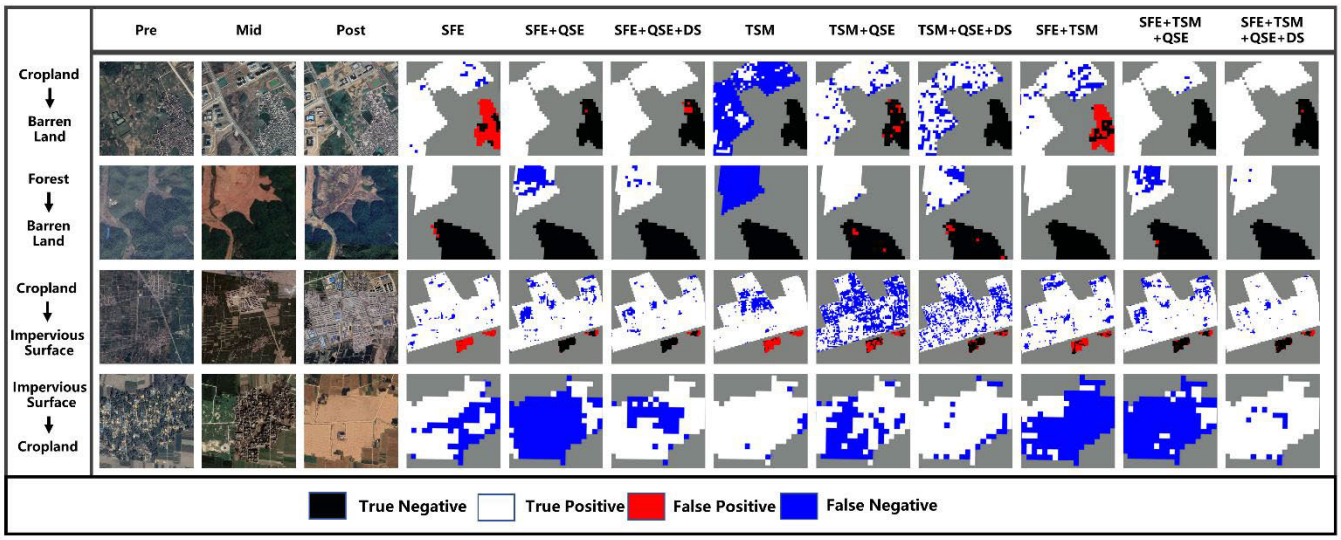


Fig. 7. Visual comparison of ablation results for TC-SSFN. QSE corrected SFE's false positives and reduced TSM's missed detections, yet slightly weakened interclass contrast. DS further enhanced feature discriminability and delivered the highest performance.

TABLE III
ACCURACY COMPARISON ACROSS THREE LAND COVER TYPES

Method	Cropland	Forest	Impervious Surface
SFE	64.92%	97.11%	39.60%
SFE+QSE	96.76%	99.58%	91.62%
TSM	5.60%	94.63%	79.00%
TSM+QSE	67.97%	99.89%	91.71%

TABLE IV
COMPUTATIONAL COST AND MEMORY USAGE OF TSM MODULE

Segmentation Granularity	Model Parameters(M)	Forward Pass Time (s)	Peak GPU Memory (MB)
30	7.40	1.2819	3438.51
90	3.38	1.5509	1194.68
180	2.39	0.9863	625.78
360	1.92	0.8081	342.61

low accuracy in cropland (5.60%) due to strong phenological effects, but the addition of QSE raised it to 67.97%, alongside marked improvements in forest (from 94.63% to 99.89%) and impervious surfaces (from 79.00% to 91.71%).

These results confirmed that the QSE module effectively mitigated the adverse effects of seasonal variations on both spatial and temporal representations. By aligning features across identical seasonal phases, QSE enabled the model to focus on genuine land cover changes rather than periodic vegetation dynamics. The improvements observed in cropland, where spectral variability was most pronounced, particularly highlighted the stability advantage conferred by seasonal differencing. Notably, even with QSE's enhancement, the temporal branch's accuracy on cropland remained substantially lower than that of the spatial branch, underscoring the inherent challenge of modeling unstable interannual dynamics and the relative robustness of spatial representations for such land cover. Moreover, the consistent gains in forest and impervious surface areas indicated that QSE not only enhanced robustness in seasonally sensitive regions but also promoted overall feature consistency across diverse land cover types.

TABLE V
COMPARISON OF CLUSTERING METRICS

Module	DS	Metric	Mean(0)	Mean(1)	p-value	Cohen's d	Interpretation
SFE	√	Spatial Entropy	1.9958	2.7537	<0.001	-0.988	Large
		Avg Nearest Neighbor	1.1257	1.3200	<0.001	-0.709	Medium
		Median Nearest Neighbor	0.1481	0.7823	<0.001	-1.891	Large
		Avg NN(Excl. Outliers)	0.1806	1.0700	<0.001	-2.074	Large
		DBSCAN Noise Ratio	0.2729	0.6185	<0.001	-1.673	Large
	×	Spatial Entropy	3.5526	3.5467	<0.001	0.083	Negligible
		Avg Nearest Neighbor	1.4196	1.4172	0.3624	0.013	Negligible
		Median Nearest Neighbor	1.0540	1.0643	0.0013	-0.047	Negligible
		Avg NN(Excl. Outliers)	1.4196	1.4172	0.3624	0.013	Negligible
		DBSCAN Noise Ratio	0.9979	0.9975	0.1049	0.024	Negligible
TSM	√	Spatial Entropy	3.0557	3.5727	<0.001	-1.001	Large
		Avg Nearest Neighbor	1.4505	2.9353	<0.001	-1.606	Large
		Median Nearest Neighbor	0.7828	2.1205	<0.001	-1.442	Large
		Avg NN(Excl. Outliers)	1.2243	2.911	<0.001	-1.616	Large
		DBSCAN Noise Ratio	0.5646	0.9587	<0.001	-1.397	Large
	×	Spatial Entropy	3.6032	3.5680	<0.001	0.226	Small
		Avg Nearest Neighbor	4.5960	4.4214	<0.001	0.188	Negligible
		Median Nearest Neighbor	3.296	3.0417	<0.001	0.259	Small
		Avg NN(Excl. Outliers)	4.5960	4.4192	<0.001	0.190	Negligible
		DBSCAN Noise Ratio	0.9998	0.998	<0.001	0.088	Negligible

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3) Impact of Temporal Segmentation Granularity: Our analysis showed that the optimal temporal granularity depended on both the intrinsic seasonal cycles of land cover types and the nature of the changes. Short segments (30 days) captured subtle, short-term variations but were prone to oversensitivity to seasonal fluctuations, particularly in croplands and paddy fields with high phenological variability. In contrast, long segments (180 or 360 days) often overlook rapid or intermediate changes, such as temporary urbanization or abrupt farmland-to-impervious surfaces' conversions, leading to missed or blurred detection of change areas. A 90-day segmentation, corresponding to a quarter-based temporal patching strategy, effectively aligned with natural seasonal cycles, enabling the model to capture both gradual seasonal variations and abrupt land cover changes while reducing misclassifications caused by pseudochanges.

Notably, in the PatchTST-based temporal feature extraction within the TSM module, the temporal segmentation directly determined the patch length: each patch corresponded to a continuous window of observations along the time axis that the Transformer processed as a single unit. Shorter segments generated more patches per sequence, increasing the number of forward passes through the Transformer and the overall computational cost. Similarly, in the SFE module, which applied spatial convolutions on local patches, finer temporal segmentation required processing more temporal slices through the convolutional layers, further increasing the number of operations, forward pass time, and GPU memory usage. Consequently, while finer granularity improved sensitivity to short-term or abrupt changes, it came at the expense of higher computational demands and memory consumption. Conversely, coarser segmentation reduced resource requirements but risked overlooking temporally localized variations.

Overall, the 90-day segmentation provided a practical compromise between capturing seasonal dynamics and maintaining manageable computational costs, offering sufficient temporal resolution to detect both gradual and abrupt land cover changes without incurring excessive memory or processing demands.

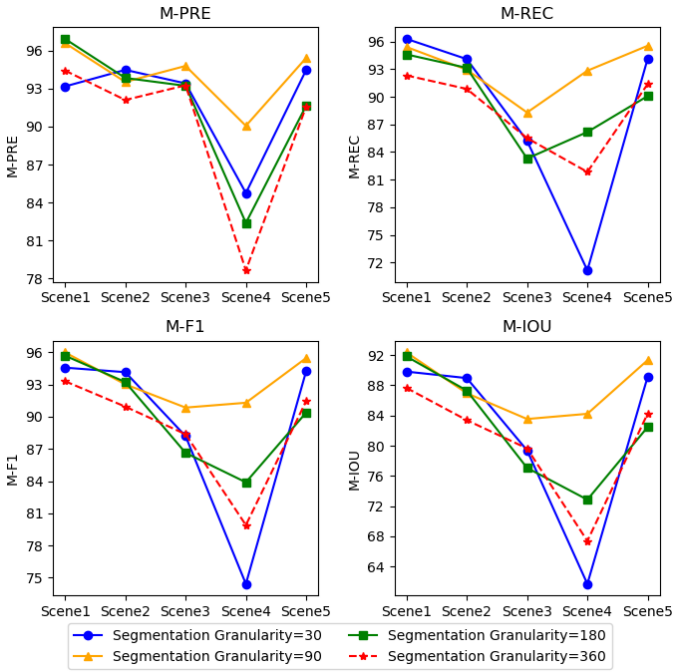


Fig. 8. Performance comparison under different temporal segmentation strategies (30, 90, 180, and 360 days). The 90-day segmentation proved more robust than finer or coarser intervals: it captured both gradual and abrupt land-cover changes while reducing pseudo-change misclassifications and maintaining manageable computational costs.

V. Discussion

A. Supervision-Induced Semantic Structuring of Feature Space

To verify the effectiveness of the proposed auxiliary supervision, we conducted a clustering-based analysis on the intermediate feature representations extracted from the SFE and TSM, both with and without DS. This analysis aimed to evaluate whether the supervision encouraged semantically meaningful and structurally discriminative features in the learned representations. Fig. 8 illustrates the impact of DS on the compactness and clustering structure of learned features. Ideally, we expected the feature representations across different time points to exhibit stronger semantic consistency and structural discriminability. For unchanged objects, their temporal features remained highly consistent, forming a compact cluster, as shown in Fig. 8(a), where most points aggregated tightly except for a few outliers. In contrast, for changed objects, the features at different time points were expected to be more distinct and dispersed, enabling better separability. As shown in Fig. 8(b), even under supervision, some degree of intracluster cohesion remained in the changed samples, which aligned with the fact that real-world changes were often gradual or intermittent. Comparatively, Fig. 8(c) and (d) presents feature structures obtained without DS. These representations appeared more scattered and failed to exhibit clear structural differences between changed and unchanged samples.

We conducted a structural analysis of features on 11 759 unchanged samples and 11 906 changed samples, covering land cover types such as buildings, vegetation, and bare soil. Several quantitative metrics were computed to assess the compactness and dispersion of features, including spatial entropy, average and median nearest neighbor distances, DBSCAN-based noise ratio, and average core-point distance. These metrics were selected to characterize feature distributions from complementary structural perspectives—namely, global organization, local compactness, and density-based clusterability.

Spatial entropy captured the overall degree of order or randomness in the feature space, indicating whether features were uniformly distributed or concentrated into structured regions. Nearest neighbor distances quantified local compactness by measuring intersample proximity, where the average distance reflected overall neighborhood tightness and the median distance provided robustness against outliers. The DBSCAN-based noise ratio evaluated density-driven clusterability by measuring the proportion of samples that failed to belong to any high-density region, thereby indicating the presence or absence of coherent semantic structures. Finally, the average core-point distance measured intracluster cohesion by

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characterizing how tightly core samples were grouped within detected dense regions. Together, these metrics provided a comprehensive and interpretable characterization of feature compactness and dispersion in the PCA-projected feature space, enabling a systematic analysis of how supervision reshaped feature organization.

Our clustering analysis of intermediate features revealed clear distinctions between models trained with and without auxiliary supervision. In the absence of supervision, the learned features exhibited weak structural organization. Both changed and unchanged samples showed similarly high spatial entropy and large nearest neighbor distances, while the DBSCAN-based noise ratio often approached 1.0—indicating that most feature points failed to form meaningful clusters. This elevated noise ratio did not indicate a limitation of the clustering algorithm but rather reflected the absence of dense and semantically coherent regions in the learned representation space. This suggested that, under unsupervised conditions, the model was unable to capture discriminative representations that reflected the semantic distinction between changed and unchanged regions.

In contrast, the introduction of auxiliary supervision led to a more semantically structured feature space. Features from unchanged regions formed tight and compact clusters, characterized by low spatial entropy, shorter nearest neighbor distances, and significantly lower noise ratios. Conversely, features from changed regions were more dispersed, with higher entropy and a greater proportion of noise points, indicating an intentional amplification of representational divergence. This asymmetric pattern was consistent with the objective of change detection: unchanged regions exhibited high intraclass consistency, whereas changed regions showed greater representational variability. These patterns were consistent across both the SFE and TSM modules, with large effect sizes and highly significant p-values, confirming that the proposed supervision strategy effectively guided the model toward learning more meaningful and class-discriminative representations.

We observed that the clustering behaviors of the two supervised modules, SFE and TSM, exhibited distinct yet complementary patterns. The SFE was designed to capture spatial difference patterns across the same seasonal phase in adjacent years, inherently emphasizing localized and abrupt changes. Under auxiliary supervision, SFE produced highly compact representations for unchanged samples, characterized by extremely low median and average nearest neighbor distances, as well as reduced spatial entropy. These results indicated strong local compactness and high global structural regularity for temporally stable regions, consistently reflected across multiple complementary metrics. In contrast, features corresponding to changed regions became substantially more dispersed, with sharp increases in entropy and DBSCAN-based noise ratio. This suggested that the supervision on SFE effectively amplified spatial deviation signals in dynamic areas while suppressing noise in temporally stable regions—thus enhancing the spatial contrast within the learned representations.

In comparison, the TSM was tailored to model seasonal dynamics by tracking the evolution of feature trajectories throughout the year. Although supervised TSM exhibited slightly higher entropy and distance values for unchanged regions compared to SFE—likely due to inherent seasonal variations—it still demonstrated clear and meaningful separation between changed and unchanged samples. The combination of moderate local compactness and preserved density-based structure indicated that TSM learned temporally consistent yet seasonally adaptable representations, rather than enforcing overly rigid clustering. Supervision enabled TSM to capture structured temporal divergence, particularly in cases of gradual or progressive change where spatial differences alone may have been insufficient. It effectively magnified representational shifts over time, reinforcing temporal discriminability.

In summary, auxiliary supervision enhanced both modules by encouraging semantically aligned feature organization according to each module's design objective. By jointly analyzing global entropy, local neighborhood structure, and density-based clusterability, the adopted metric framework provided a principled, multiscale interpretation of how supervision reshaped feature distributions. Their complementary strengths—with SFE focusing on spatial contrast and TSM capturing temporal dynamics—jointly contributed to more comprehensive, discriminative, and interpretable feature representations, ultimately improving the model's change detection performance.

B. Effect of Differencing Position on Spatial and Temporal Feature Learning

To investigate the influence of differencing strategies on the quality of learned features, we designed a set of controlled ablation experiments applied to both the spatial and temporal branches of our change detection model. Each branch explored three strategies: 1) differencing after feature extraction; 2) differencing before feature extraction; and 3) no differencing as the baseline. These experiments were conducted on five diverse scenes and evaluated using four metrics—M-PRE, M-REC, M-$F1$, and M-IOU—to reveal how differencing affected spatial and temporal feature spaces.

In the spatial branch, the strategy of applying differencing before feature extraction consistently yielded the best performance across all evaluation metrics, particularly in M-$F1$ and M-IOU. This suggested that performing differencing on raw spatial features prior to encoding could effectively suppress background redundancy and enhance structural distinctions. By highlighting change-sensitive boundaries and edges early on, the model was better guided during subsequent convolutional encoding, which tended to aggregate spatial context. In contrast, applying differencing after feature extraction led to inferior performance, likely due to the smoothing effects of convolutional layers that reduced spatial variance and obscured fine-grained structural differences, thereby introducing noise and diminishing spatial consistency. The SFE Only configuration performed moderately, implying that while spatial encoding alone could capture meaningful representations, the inclusion of a well-positioned differencing operation significantly enhanced its effectiveness.

In the temporal branch, applying differencing after feature extraction outperformed other configurations overall,

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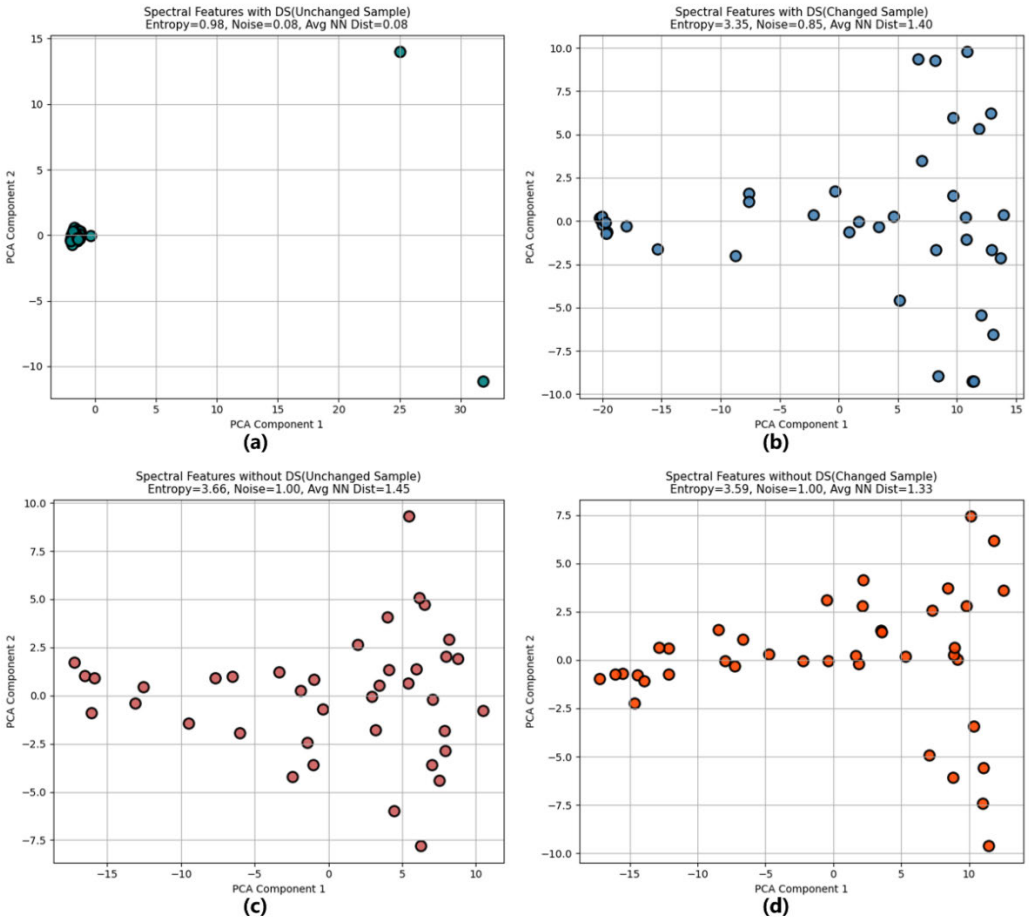


Fig. 9. PCA visualizations of spectral features under four settings. (a) Supervised + unchanged. (b) Supervised + changed. (c) Unsupervised + unchanged. (d) Unsupervised + changed. The plots revealed the effect of DS on feature compactness and cluster structure.

underscoring the importance of preserving temporal continuity during feature encoding. By first capturing intrinsic temporal patterns and subsequently computing differences in the latent feature space, the model was better equipped to isolate meaningful changes while suppressing seasonal or stochastic fluctuations. In contrast, performing differencing directly on raw time series inputs led to a significant drop in performance. This was attributed to two critical issues: 1) in cases where phenological events were delayed or advanced between years, differencing at fixed time points incorrectly detected changes where none existed, increasing the risk of false positives and 2) when both time periods exhibited smooth gradual transitions, the pointwise difference yielded near-zero values, causing genuine changes to be overlooked. These findings illustrated that early differencing disrupted the continuity of temporal signals and impaired the model's ability to capture dynamic but semantically coherent evolution patterns. Interestingly, the "TSM Only" configuration yielded performance comparable to postextraction differencing, indicating that the temporal encoder itself had a certain capacity to model cross-temporal contrast. However, the addition of differencing in the latent space further amplified discriminative signals, suggesting that structural integration of contrastive mechanisms enhanced temporal sensitivity without compromising temporal context.

The contrasting behavior between branches underscored the asymmetric nature of spatial and temporal information: spatial features benefited from early differencing to enhance local contrast, while temporal features required preservation of sequential order prior to differencing to avoid losing temporal semantics. Moreover, differencing had a stronger effect in the spatial branch, suggesting that space-based change localization was more sensitive to differencing design. In contrast, temporal differencing was more delicate and should be applied after temporal abstraction.

C. Effectiveness of Structured Temporal Modeling and Feature Learning

The consistent performance of TC-SSFN across diverse scenes indicated that structuring temporal information and guiding feature learning both played important roles in robust change detection. The proposed design stabilized temporal representations through season-aware alignment, improved feature discriminability via DS, and further enhanced robustness by integrating temporal and spatial modeling.

1) Practical Advantage of Seasonal Alignment: Our experimental results (see Table I and Fig. 6) showed that while other deep learning models achieved competitive results in

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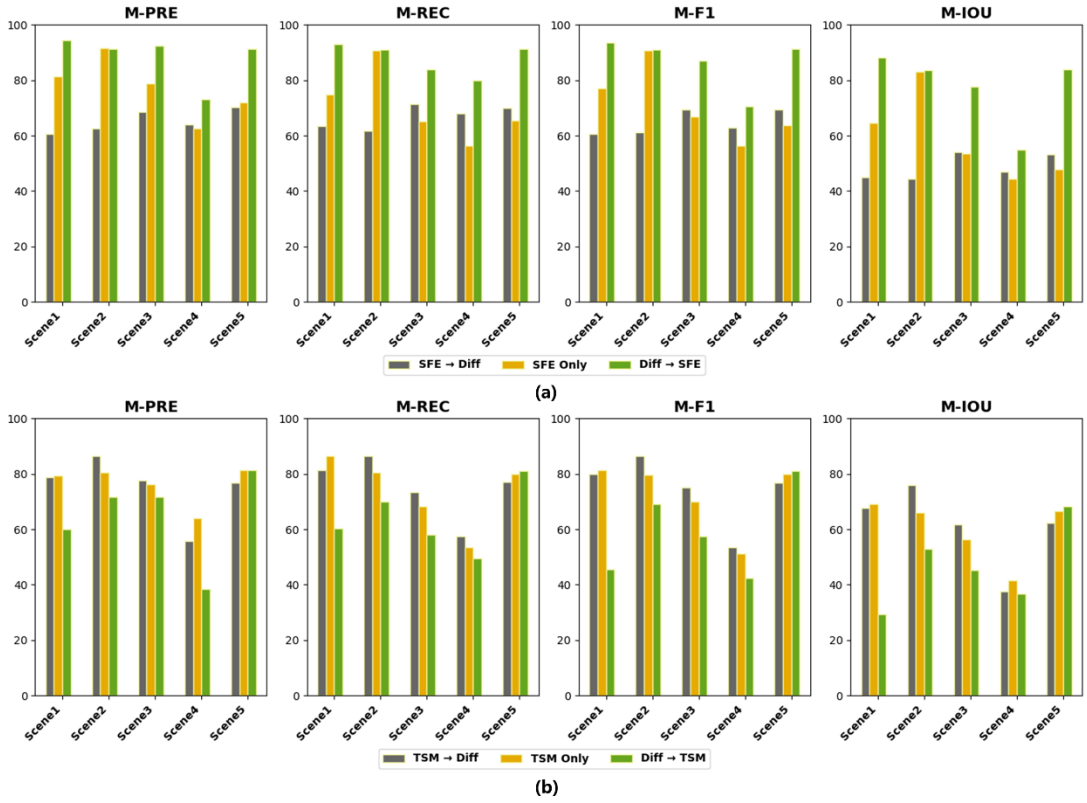


Fig. 10. Performance comparison of differencing strategies in spatial and temporal feature branches. Spatial features benefited from differencing before extraction, whereas temporal features required differencing after extraction to maintain temporal continuity.

some scenes, their performance exhibited notable variability. This observed interscene instability could be attributed, in part, to the challenge of consistently learning to separate strong seasonal patterns from true change across different geographical and phenological contexts. In contrast, TC-SSFN's QSE module, by design, addressed seasonal variation through interannual differencing of aligned quarters. The resultant stability in TC-SSFN's performance metrics indicated that this structured preprocessing of temporal signals provided a reliable foundation, reducing the model's sensitivity to scene-specific seasonal characteristics. This approach offered a pragmatic and interpretable strategy for handling a major source of noise in time series analysis.

2) DS as Direct Feature Learning Guidance: The ablation studies and feature space analysis (see Fig. 9) demonstrated that the parameter-free DS module significantly enhanced feature discriminability. This outcome highlighted an alternative role for auxiliary supervision. Unlike paradigms that introduced additional classification tasks at intermediate layers, our DS module acted as a direct regularizer on the feature representation itself. It encoded a simple, task-relevant logic—features of unchanged areas should be temporally consistent—directly into the learning objective. The resulting feature structure (compact clusters for unchanged, dispersed points for changed) indicated that this form of supervision effectively guided the network to internalize a core principle of change detection, contributing to both performance and the interpretability of the learned features.

3) Complementary Synthesis of Methodological Strengths: The effectiveness of the CCDC algorithm highlighted the importance of temporal information for change detection, particularly its strategy of characterizing temporal profiles and identifying changes based on deviations from expected temporal patterns. Inspired by this idea, our framework adopted a similar comparison-based perspective and further extended it to a season-aware setting by aligning and contrasting corresponding seasonal phases, which helped reduce the influence of temporal noise and irregular fluctuations in the time series. At the same time, CCDC emphasized pixelwise temporal modeling, while convolutional networks were effective in capturing spatial–semantic structures. These two perspectives naturally complemented each other. Building upon this observation, TC-SSFN integrated structured temporal comparison with spatial feature learning, enabling joint modeling of temporal dynamics and spatial context, and resulting in more robust and reliable change detection across diverse scenes.

D. Limitations and Future Work

The TC-SSFN framework, while demonstrating strong performance, was subject to certain limitations inherent to its design. First, its core architecture assumed a regularized temporal input structure for subsequent quarter-based alignment and modeling. This necessitated interpolation to form a consistent series, making the method dependent on the underlying availability and quality of optical observations. Persistent cloud cover or lengthy data gaps, despite gap-filling,

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could affect the integrity of the constructed sequences. Second, the model in its current form was designed for binary change detection. While this addressed a fundamental question, it did not delineate the specific land-cover transition types, which were often critical for downstream applications. These limitations defined clear and valuable trajectories for future research.

1) Robust Multisource Time Series Construction: To reduce reliance on a single optical data source, future work will focus on integrating data from multiple sensors. The goal is not to eliminate interpolation but to enhance the quality and temporal density of the foundational input series, thereby increasing the model's resilience to cloud contamination and data outages.

2) Toward Semantic Change Understanding: A primary focus will be extending the framework from binary to semantic change detection. This will involve generalizing the DS mechanism and the feature disentanglement objectives to manage multiple "from-to" change classes. The challenge lies in designing label guidance that could teach the network to distinguish between spectrally similar but semantically different transitions.

VI. Conclusion

We proposed a robust change detection network (TC-SSFN) that integrated SFE, temporal series modeling (TSM), QSE, and DS. Across five representative scenes, TC-SSFN achieved an average $F1$ score of 93.33% and an IOU of 87.69%, outperforming the comparison methods in overall accuracy and robustness. Experimental results showed that QSE effectively mitigated seasonal pseudochanges, stabilizing predictions in phenologically dynamic regions. The DS module structured the feature space, generating compact clusters for unchanged regions and dispersed features for changed areas, thereby enhancing discriminability. Quarter-based temporal segmentation balanced sensitivity to both gradual and abrupt changes with computational efficiency, while differencing strategies indicated that early spatial differencing combined with post-encoding temporal differencing best preserved meaningful change patterns. Overall, the proposed approach improved both robustness and interpretability in detecting land cover changes from seasonal remote sensing time series. Future work will explore semantic-level land cover transitions and the integration of multisource remote sensing to further enhance change detection performance.

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